

# Zach Lawrence

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## Research Interests

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AI for science, foundations of machine learning, scientific computing, and time series analysis.

## Education

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*Ph.D Computer Science* — **University of Wisconsin, Madison**  
September 2026 – June 2031

*B.S. Computer Science* — **University of California, San Diego**  
Minors in Physics and Mathematics  
September 2022 – June 2026  
GPA: 3.87/4.0

## Publications

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Cao, Y, Lai, S, Huang, H, Zhang, Y, **Lawrence, Z**, Bhakta, R, Thomas, I, Cao, M, Tsai, C, Zhou, Z, Zhao, Y, Liu, H, Marinoni, A, Arefiev, A, Yu, R. SimulCost: A Cost-Aware Benchmark and Toolkit for Automating Physics Simulations with LLMs. International Conference on Machine Learning (ICML) 2025.

**Lawrence, Z**, Yao, J, Qin, C. Deep Learning for Modeling and Dispatching Hybrid Wind Farm Power Generation. International Conference on Data Mining Workshops (ICDMW) 2025. In press, [arXiv:2512.00728](https://arxiv.org/abs/2512.00728) [🔗](#).

## Presentations

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**Lawrence, Z**, Yao, J. "Deep Learning for Modeling and Dispatching Hybrid Wind Farm Power Generation". International Conference on Data Mining Workshops (ICDMW) 2025, 12 Nov. 2025, Washington D.C.

## Posters

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**Lawrence, Z**, Yao, J. "Deep Learning for Strategic Power Generation for Hybrid Wind Farms". Summer Undergraduate Research Symposium, 1 Aug. 2025, Vancouver, WA.

Yao, J, **Lawrence, Z**. "Atmospheric Data-Driven Time Series Modeling of Wind Power Generation". Summer Undergraduate Research Symposium, 1 Aug. 2025, Vancouver, WA.

**Lawrence, Z**, Wrye, G, Panthi, A, Rai, R. "Mangrove Monitoring". E4E REU Symposium, Aug. 2024, Escondido, CA.

## Experience

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**Washington State University, Vancouver**  
NSF-Funded Undergraduate Researcher (REU)  
June 2025 - August 2025

—Designed and evaluated classical probabilistic and deep learning models for estimating power generation and optimizing power storage strategies for wind farms with integrated energy storage under Dr. Chris Qin.

—For power generation estimation, developed a recurrent neural network (RNN) architecture based on NQF-RNN, a probabilistic forecasting architecture, which outperformed benchmarks by 9% RMSE on historical data.

—For power dispatch, developed a novel RNN with unsupervised learning designed to maximize energy value, which reduced cost of energy compared to conventional baseload dispatch by 32.3% COVE on historical data.

—Full paper accepted and presented at REU workshop at 25th International Conference on Data Mining (ICDM).

**Rose Yu Spatiotemporal Lab**  
Undergraduate Researcher  
University of California, San Diego  
April 2025 - Present

—Developed Bayesian active learning framework for deep learning surrogate models, designed to approximate high-fidelity turbulent transport model for nuclear fusion simulations, under Dr. Rose Yu and Dr. Yadi Cao and in collaboration with General Atomics (*paper in progress*).

—Engineered a Docker and Kubernetes-based framework for automatic scheduling of simulation jobs on the NSF-funded Nautilus computing cluster.

—Designed novel acquisition functions for Bayesian active learning, dubbed regression DIRECT (rDIRECT) and stratified expected information gain (sEIG), which extend state-of-the-art data acquisition methods.

### Engineers for Exploration

Undergraduate Researcher on Acoustic Species Identification Team

University of California, San Diego

April 2025 - Present

—Designed and trained a lightweight CNN based on the MobileNetV2 architecture for bird call classification with advising by Dr. Ryan Kastner and Dr. Curt Schurgers, (*paper in progress*).

—Profiled model performance using various post-training quantization schemes, and deployed to an embedded device using code generation tools.

### Engineers for Exploration

Project Lead on Mangrove Monitoring Team

University of California, San Diego

April 2023 - June 2025

—Led a team of undergraduate developers for a web platform that identifies mangroves from aerial imagery using a semantic segmentation model under advising from Dr. Ryan Kastner and Dr. Curt Schurgers.

—Introduced continuous user-facing feedback on image processing by implementing WebSocket connections between an ExpressJS-based central web server, a Flask image classification server, and a ReactJS frontend.

*NSF-funded Undergraduate Researcher (REU)*

June 2024 - August 2024

—Designed, provisioned, and deployed AWS infrastructure using EC2 and Fargate computing instances with data storage in an S3 bucket, orchestrated by an ECS Cluster, all designed to adhere to least trust principles.

—Deployed semantic segmentation model onto AWS-hosted Flask server for end-to-end web pipeline.

## Projects

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### Understanding LSTM Training Dynamics using Dynamical Systems Methods

University of California, San Diego

February 2025 - Present

—Developed deep learning methodology for probabilistic forecasting of fractional Brownian motion under Dr. Massimiliano Di Ventra and Dr. Yuanhang Zhang (*paper in progress*).

—Used dynamical systems methods for analyzing autocorrelation and cross-correlation of LSTM memory cells towards an explanation of stochasticity and its impact on training dynamics.

## Selected Coursework

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|-------------------------|---------------------------------|------------------------------------|
| Algorithm Design        | Electricity and Magnetism       | Mathematical Modeling I            |
| Computer Graphics       | Optics                          | Statistical Methods                |
| Software Engineering    | Quantum Mechanics               | Classical Machine Learning         |
| Compiler Construction   | Data Structures                 | Deep Learning                      |
| Computer Architecture   | Functional Programming          | Neuromorphic Computing             |
| Cryptography            | Computational Stochastics       | † Quantum Computing                |
| Discrete Mathematics    | Linear Algebra                  | Embedded Systems Project           |
| Computational Physics I | Calculus I, II, III             |                                    |
| Classical Mechanics     | Vector Calculus                 |                                    |
| Fluid Mechanics         | Intro to Differential Equations | † <i>Graduate Level Coursework</i> |